

The Curious Case of Apple and Its Intelligence

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Agenda

- Apple Intelligence Basics
- Use & (Ab)Use of Apple Intelligence
- Apple Intelligence Forensic Artefacts
- Challenges and Tool Release

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- The views and opinions presented in this research are personal and do not represent any employer or affiliated organization.
- This is an independent research conducted outside of professional employment responsibilities.
- For educational and security research purposes only.

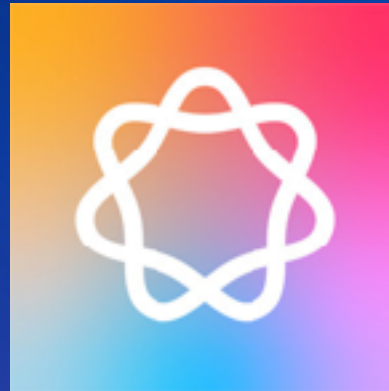


Siri Reborn:

The dawn of Apple Intelligence and Siri AI



Siri



Apple
Intelligence



Siri AI

Apple Intelligence Timeline



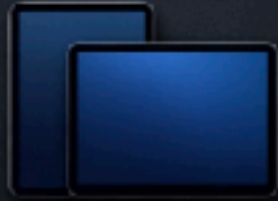
Prerequisites of Apple Intelligence

iOS 18.1 or later



iPhone 15 Pro,
iPhone 15 Pro Max,
iPhone 15 models or later

iPadOS 18.1 or later



iPad mini (A17 Pro),
iPad models with M1 or later

macOS Sequoia 15.1 or later



Mac with M1 or later

visionOS 2.4 or later



M2 chip and later

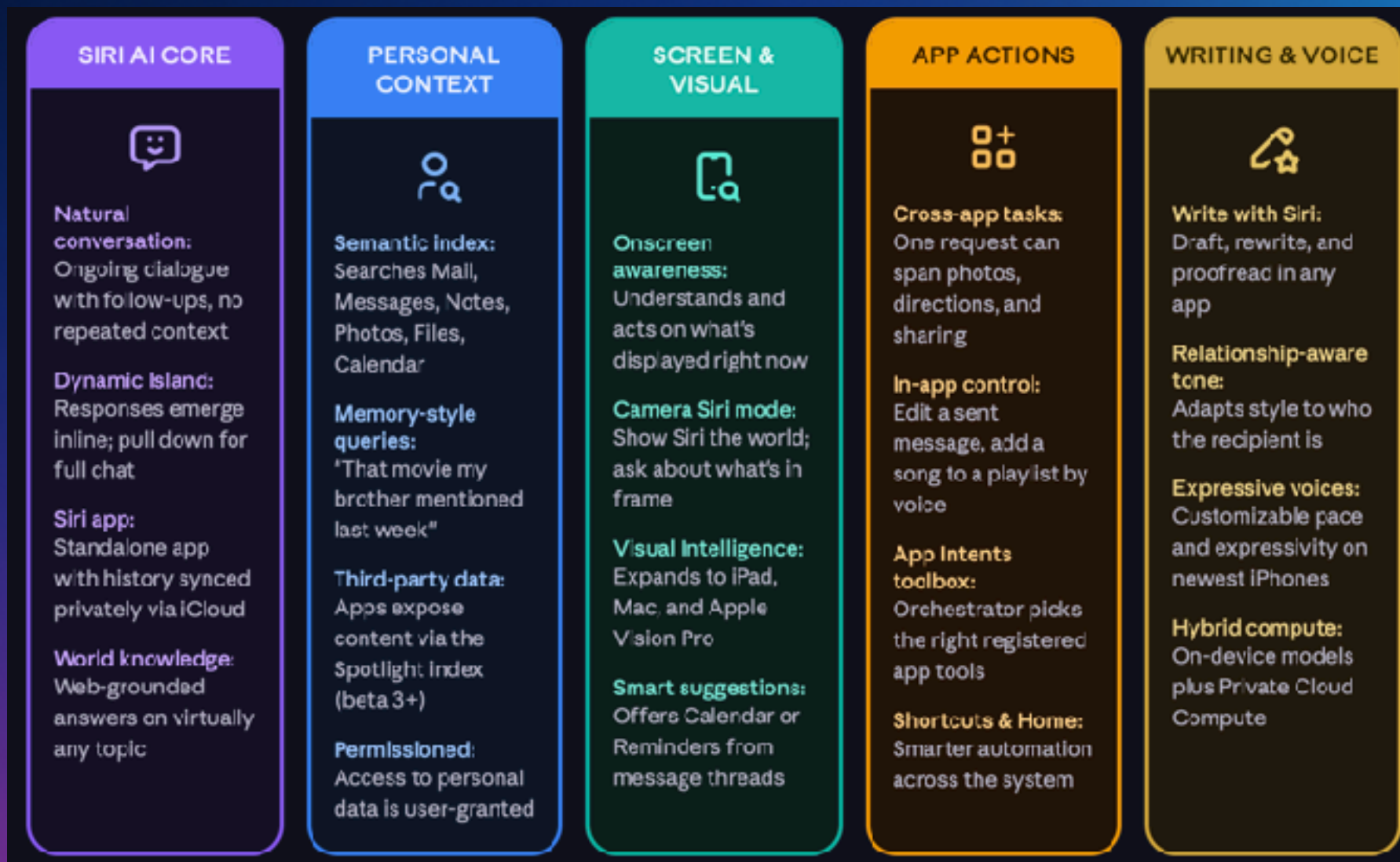
watchOS 11 or later



- Apple Watch Series 6 or later
- All Apple Watch Ultra models
- Apple Watch SE 2 or later

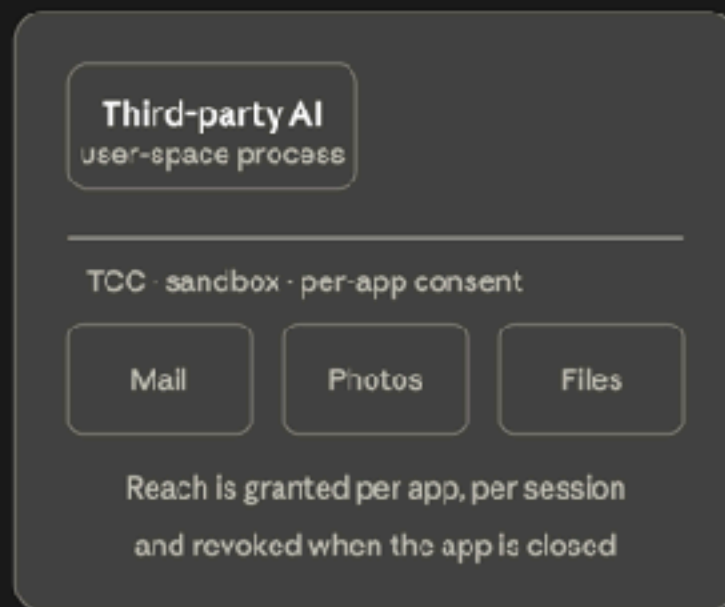
Requires an Apple Intelligence-enabled iPhone nearby

Apple Intelligence Capabilities



Why Apple Intelligence

App-scoped, user-invoked



OS-scoped, ambient



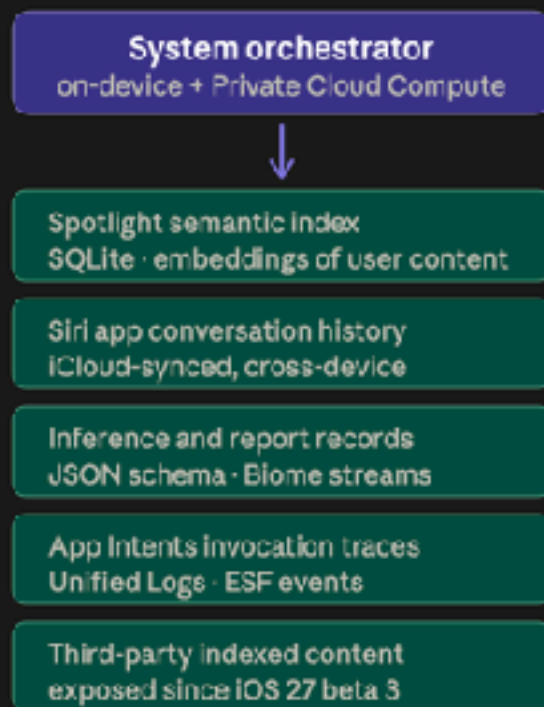
The boundary moved

Third-party AI reaches your data only when you hand it over, one app at a time.
 Apple Intelligence starts with an index of everything and a toolbox of actions, brokered by a system process rather than granted by a consent prompt.

Every path on the right writes to disk.

Most of what it writes is undocumented as of now — research work is in progress.

Apple Intelligence writes



Commercial DFIR tooling sees



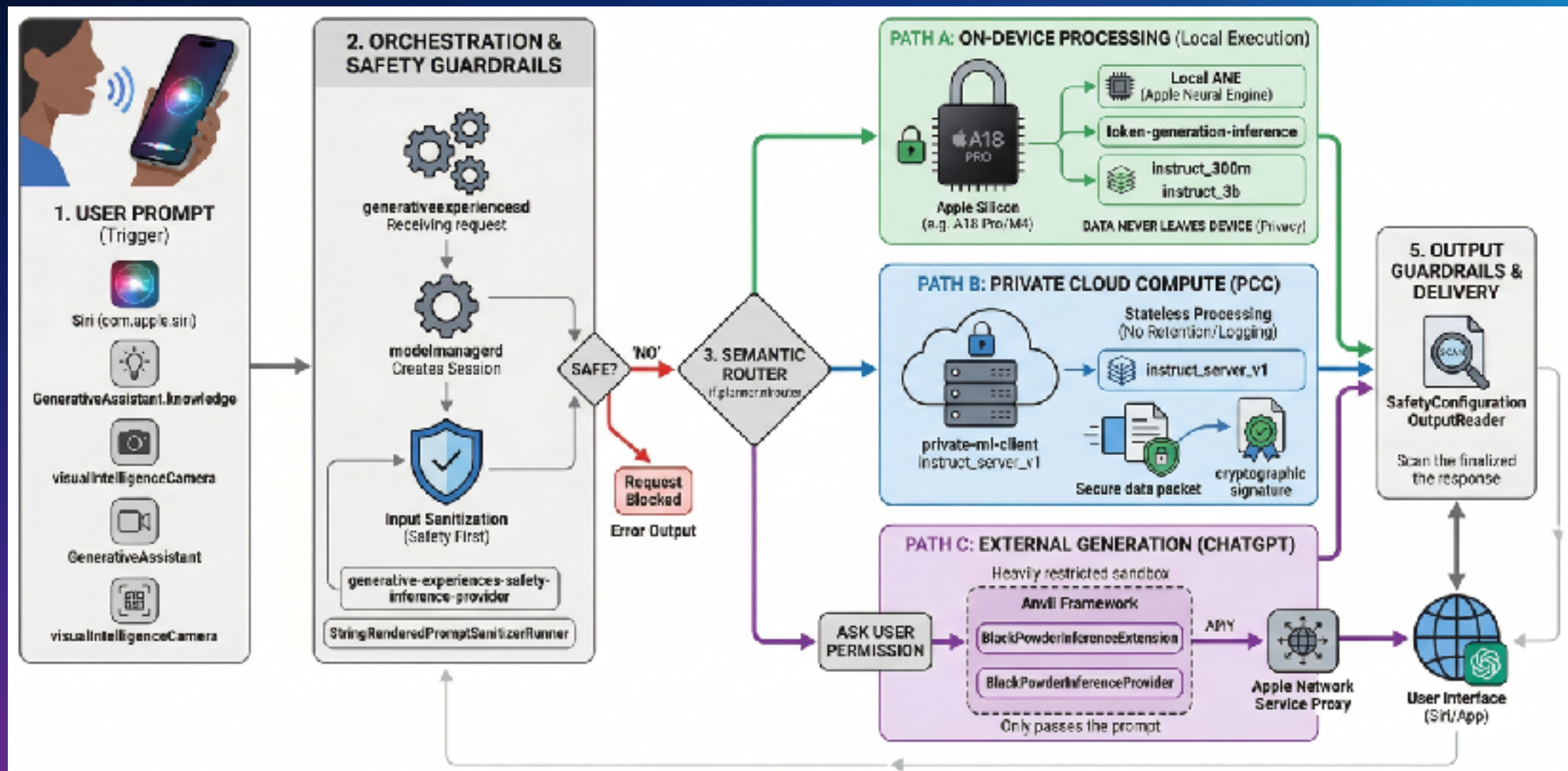
The artifacts are on disk. The tools report clean.

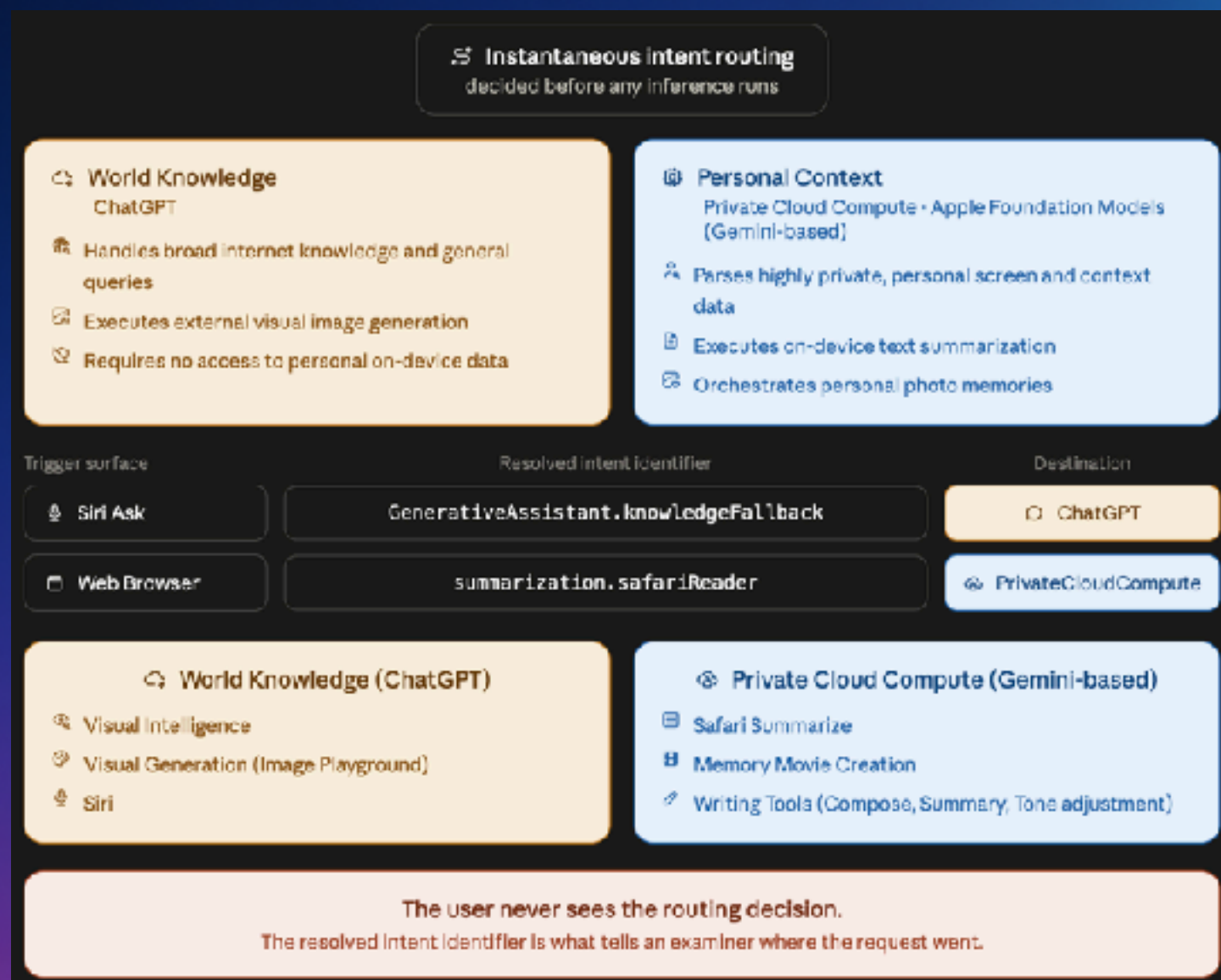
This is a parser coverage gap, not an evidence gap — and it is closable.

Artifact classes shown reflect iOS 27 / macOS 27 beta observations; research in progress

Under the Hood of Apple Intelligence

The Journey of a Prompt via Apple Intelligence





Use and (Ab)Use of Apple Intelligence



Attacker's Perspective

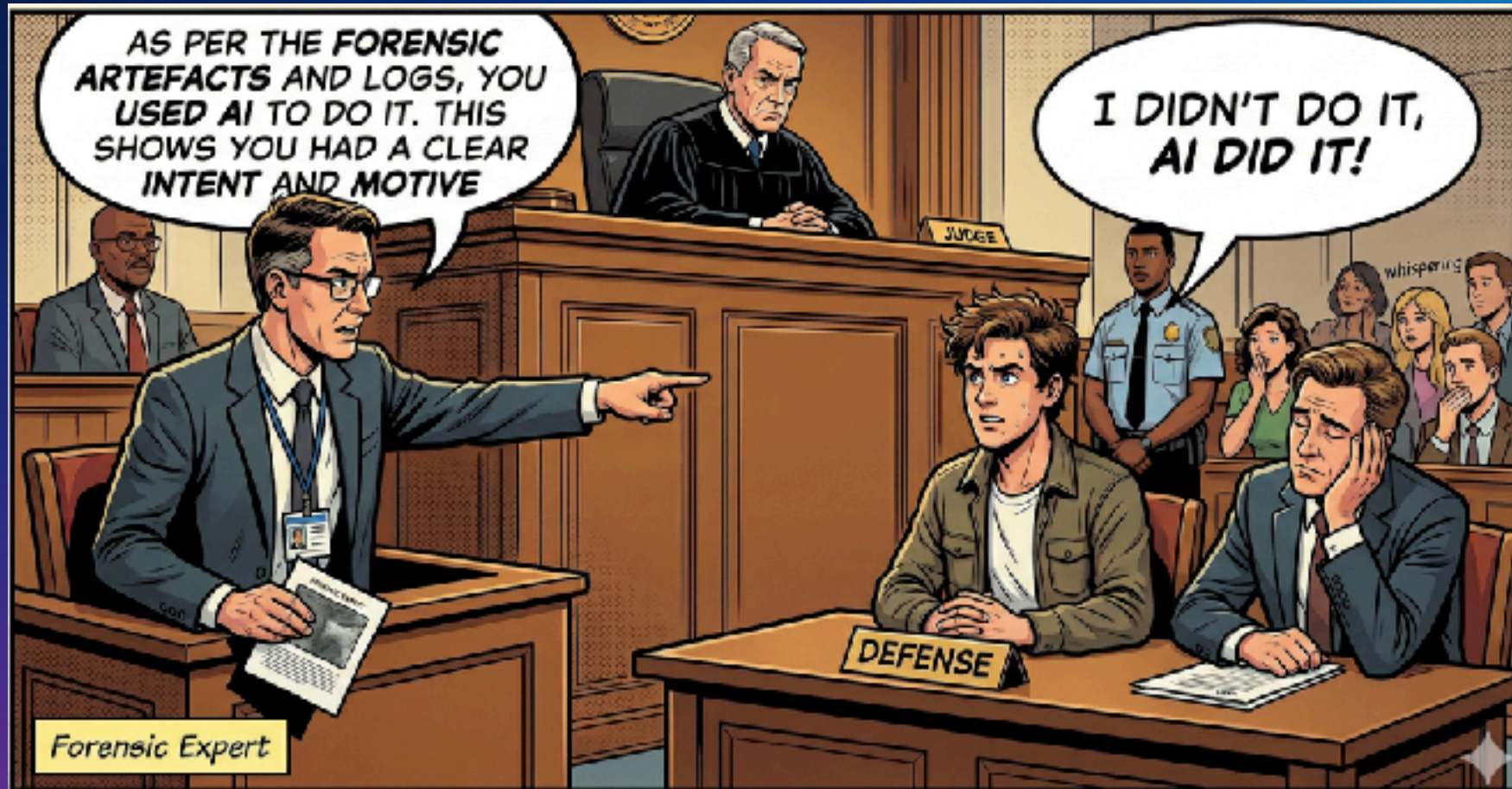
Defender's Perspective



Defender's Perspective

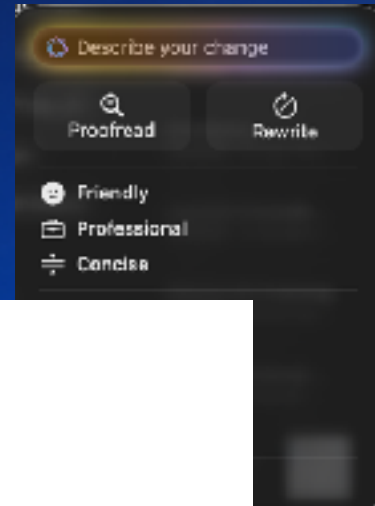
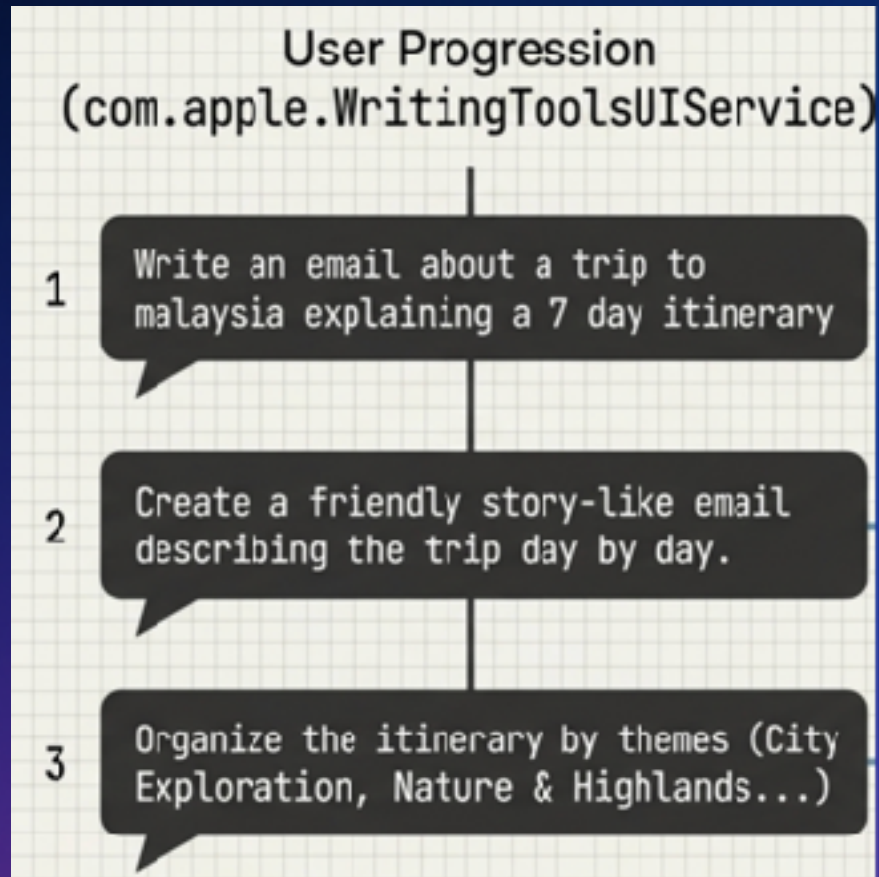


Defender's Perspective



Apple Intelligence Forensic Artefacts

Text Generation & Modification



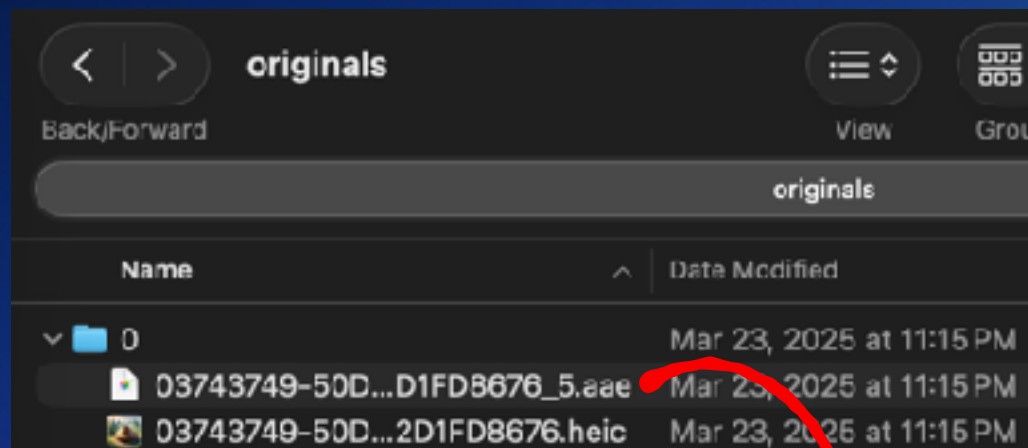
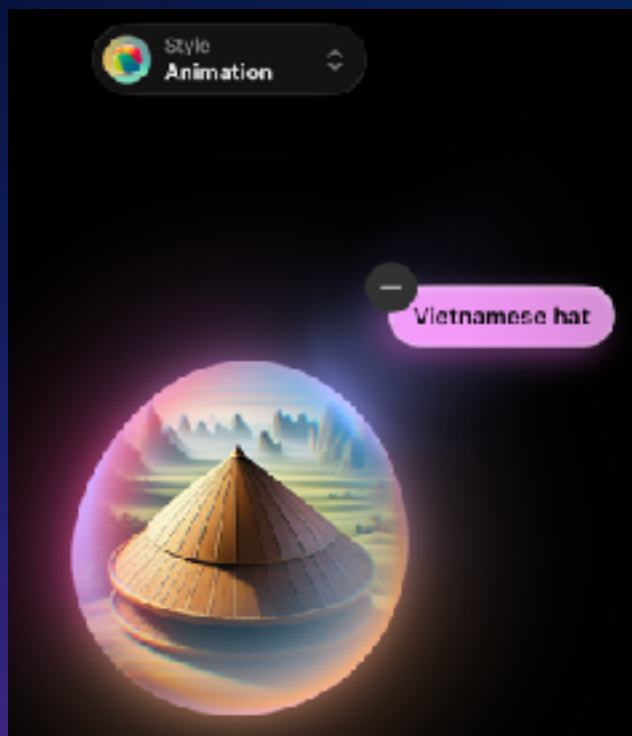
Base Model:
com.apple.fm.language.
instruct_server_v1.base

Raw Input

7-day Malaysia itinerary includes Kuala Lumpur, Cameron Highlands, Penang, and Langkawi, with a mix of sightseeing, cultural experiences, and relaxation.

- City Exploration in Kuala Lumpur
- Nature in Cameron Highlands
- Heritage in Penang
- Relaxation in Langkawi

Image Playground



```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<plist version="1.0">
<dict>
  <key>adjustmentBaseVersion</key>
  <integer>0</integer>
  <key>adjustmentData</key>
  <data>
    YnBsaW80MDRBAQ1haz5ncrYkaiWudH0i6wzBBAYVchIv6YD000YHUI8w0qoICotVdnEs
    qWYYZh58aXRuZXNeVn1ldG5lLmVwZS5S8eYYS0800YXN0erx180YPMFuan1ldG5lbnQl
    FhodJGcclZU+TUSRVloAAAAAAAAABAAQAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAZ=
  </data>
  <key>adjustmentFormatIdentifier</key>
  <string>com.apple.GenerativePlayground</string>
  <key>adjustmentFormatVersion</key>
  <string>6.0</string>
  <key>adjustmentRenderTypes</key>
  <integer>0</integer>
  <key>adjustmentTimestamp</key>
  <date>2025-03-15T13:41:28Z</date>
</dict>
</plist>
```

Output

```
bplist00N..[ingredientse..N..VpromptN..R_00.
.UvalueXentities^Vietnamese hat N
.UstyleN..Yanimation.....$'*/5>MNQWZ.....
```

Image Playground Exif data



```
ExifTool Version Number      : 13.18
File Name                    : 03743749-50DC-485B-9C0A-D1F2D1FD8676.heic
Directory                   : /Users/bhargavrathe/Library/Photos/Libraries/Application/com.apple.Gen
                             -0000-0000-000000000001.photoslibrary/originals/0
File Size                    : 136 kB
File Modification Date/Time  : 2025:03:23 23:15:17+05:30
File Access Date/Time       : 2026:04:07 21:42:34+05:30
File Inode Change Date/Time  : 2025:03:23 23:15:19+05:30
File Permissions             : -rw-----
File Type                   : HEIC
File Type Extension         : heic
MIME Type                   : image/heic
Major Brand                 : High Efficiency Image Format HEVC still image (.HEIC)
Minor Version               : 0.0.0
Compatible Brands           : mif1, m1af, MiHB, heic
Handler Type                : Picture
Primary Item Reference       : 5
Meta Image Size             : 1024x1024
XMP Toolkit                 : XMP Core 6.0.0
Credit                     : Apple Image Playground
Digital Source Type         : http://cv.ipc.org/newscode/digitalsourcetype/trainedAlgorithmicMedia
Profile CMM Type            : Apple Computer Inc.
Profile Version             : 4.0.0
Profile Class               : Display Device Profile
Color Space Data            : RGB
Profile Connection Space    : XYZ
Profile Date Time           : 2022:01:01 00:00:00
Profile File Signature      : acsp
Primary Platform            : Apple Computer Inc.
```

Image Playground Extended Attributes



`com.apple.assetsd.creatorBundleID:`

```
00000000 63 6F 6D 2E 61 70 70 6C 65 2E 47 65 6E 65 72 61
00000010 74 69 76 65 50 6C 61 79 67 72 6F 75 6E 64 41 70
00000020 70 2E 4D 65 73 73 61 67 65 73 45 78 74 65 6E 73
00000030 69 6F 6E
00000033
```

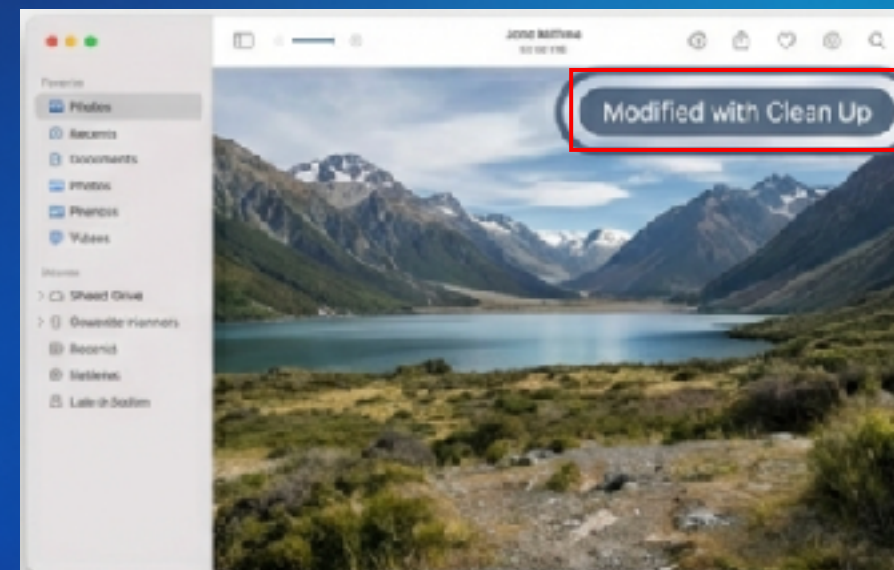
`com.apple.General
ativePlaygroundAp
p.MessagesExtens
ion`

`com.apple.assetsd.importedByDisplayName:`

```
00000000 50 6C 61 79 67 72 6F 75 6E 64
0000000A
```

`Playground`

Image Cleanup



Before Cleanup (Original)

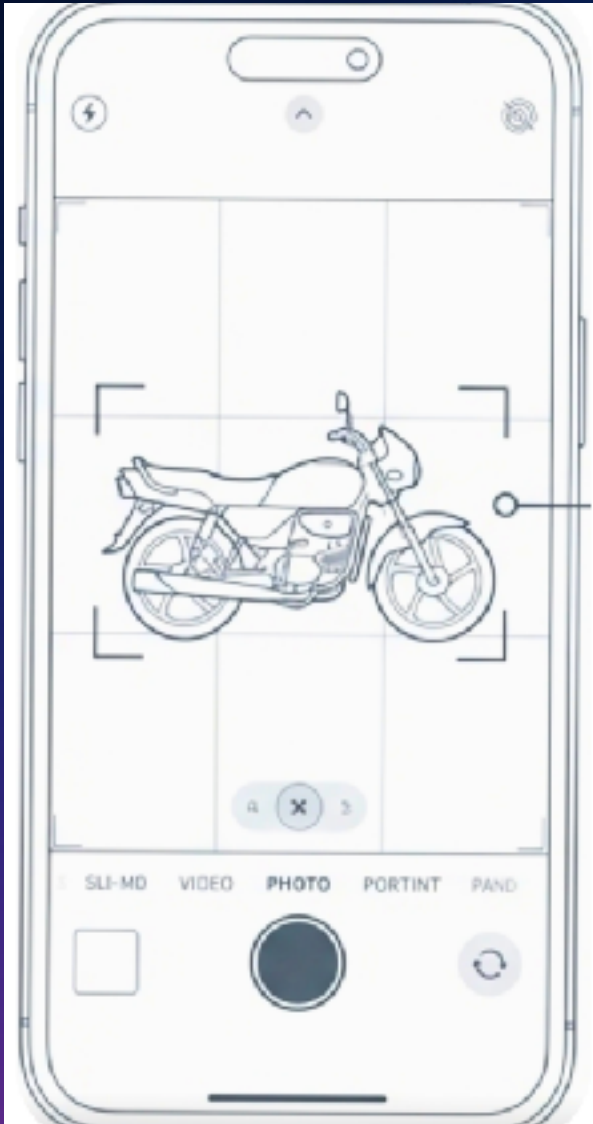
```
Exif: Device Model: iPhone 15 Pro
Exif: Location Data: +13.7563, +100.5018
xattr: com.apple.assetsd.UJID: 7365423A...
xattr: com.apple.quarantine: sharingd
```

After Cleanup (Modified)

```
Exif: Device Model: [STRIPPED]
Exif: Location Data: [STRIPPED]
xattr: com.apple.assetsd.* [FLUSHED]
xattr: com.apple.quarantine: Photos
```

```
Exif.Image.Credit : Apple
Exif.Photo.DigitalSourceType :
http://cv.iptc.org/newscodes/digitalsource
type/compositeWithTrainedAlgorithmicMedia
```

Visual Intelligence



User: Identify the object

```
{
  "prompt": "Identify the object",
  "response": "{\\"content\\":The object is a Hero Honda Splendor
motorcycle, a commuter bike known for its durability...\\"},
  "It has motocyced identifies in any connouality...\\"",
  "clientIdentifier": "com.apple.siri",
  "useCase": "GenerativeAssistant.visualIntelligenceCamera"
}
```

Visual Intelligence

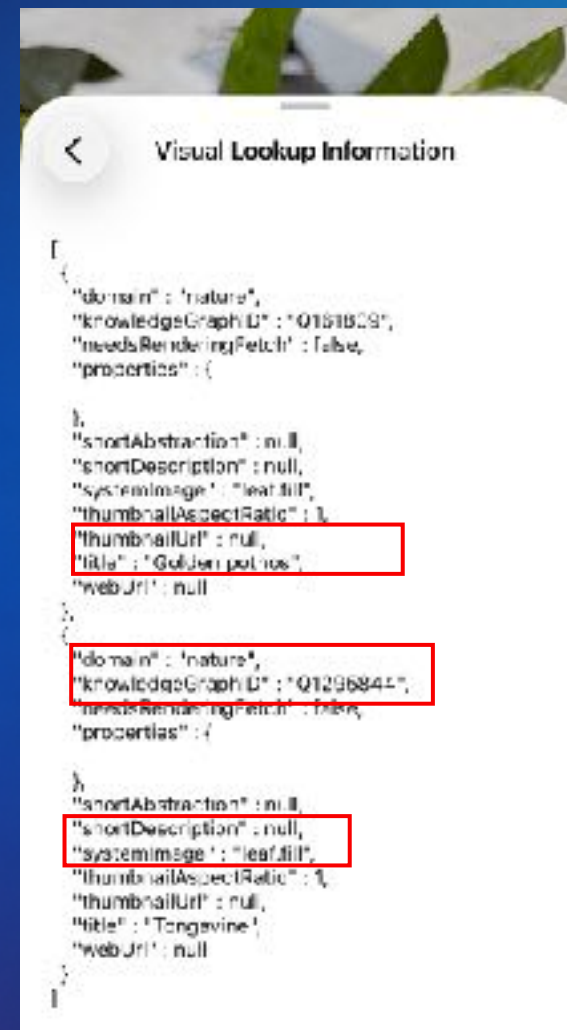


User:

Is this a snake plant?

Apple Intelligence:

No... Golden pothos has heart-shaped leaves, while snake plants have tall, upright, sword-like leaves.



Cross Device Artifacts



Image Wand
Tracing Sketch to
Image Creations
originating in Notes
iOS but appearing on
macOS



Visual Intelligence
Identifies images
captured to answer
queries or auto-create
Calendar events,
leaving sync traces



Apple Intelligence Data Remanence

Local Data Retention

- 15 min to ~ 7 days
- Sometimes ~1 year

Artifact State

- Unencrypted
 - Apple Intelligence Report (AIR)
- Encrypted
 - Private Cloud Compute (PCC)
 - Biomes

Artifact Data Format

- Apple Proprietary Files
- SEGB
 - NSKeyedArchiver
 - MacBinary

Data Storage Location

- Some Data is Ephemeral
 - Volatile Memory
- Most Data stored locally
- Some Data is stored on cross devices connected with a Single Apple account

Summary: Apple Intelligence Forensic Artefacts

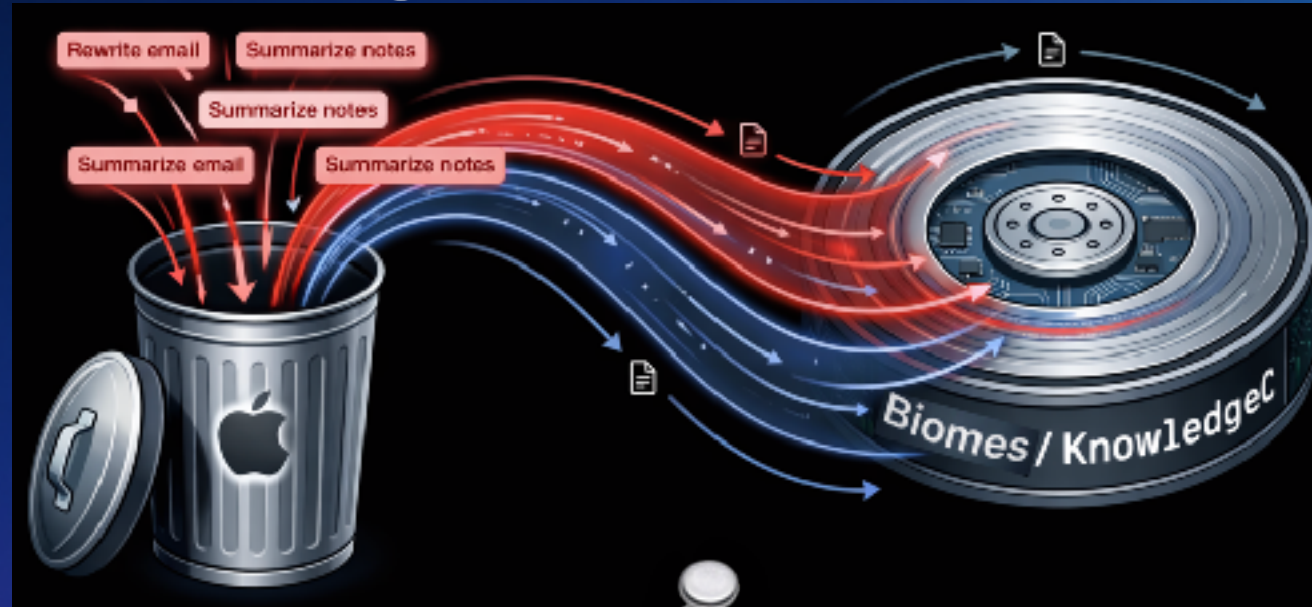
Primary Artefacts

- Apple Intelligence Report (AIR)
- Apple Unified Logs (AUL)
- Biomes
 - PCC & Siri
- KnowledgeC
- Extended Attributes
 - MDLS (Metadata Listing)
 - XATTR (Extended Attributes)

Other Artefacts

- EXIF Metadata
 - Not stored in the filesystem
- Apple Endpoint Security Framework (ESF Logs)
 - Ingested into EDR/XDR (only macOS)
- Sysdiagnose Logs
 - Requires manual trigger to generate
- Apple Account/ iCloud Artefacts

Apple Intelligence Data Remanence



Data Remanence

When a user deletes an AI generated Photo or Note, the corresponding AI metadata and index **entries are not removed.**

Hidden Artifacts

Traces remain deeply embedded in **encrypted and unencrypted** localized **Biome** streams (**even from other Apple Devices!**).

Investigative Goldmine

These lingering traces allows investigators to **reconstruct timelines** and **prove intent** despite data deletion.

Anatomy of an Apple Intelligence AI Request

The User Intent

The raw natural language query provided by the user.

The OS Context

How the operating system categorizes the action.

```
{
  "timestamp": 1772860970.758718,
  "identifier": "7D54929D-3016-4A13-91B2-96A705E54F1F",
  "prompt": "Show me the ingredients used for baking omelette",
  "response": "{\n  \"content\": \"Basic ingredients for a baked omelette include:\n\n- Eggs\n- Mlk or cream\n- Salt and pepper\n- Butter or oil (for greasing)\n  add-ins:\n\n- Cheese (cheddar, mozzarella, feta, etc.)\n- Vegetables (onions, bell peppers, spinach, mushrooms, tomatoes)\n- Cooked meats (ham, bacon, sausage, chicken)\n- Herbs (parsley, chives, basil)\n\nWhisk the eggs with milk, season, mix in fillings, and bake until set.\",\n  \"summary\": \"List of common ingredients for a baked omelette with optional add-ins.\",\n  \"isRequestingUserInput\": false\n}",
  "model": "com.apple.openai.chatgpt",
  "useCase": "GenerativeAssistant.KnowledgeFallback",
  "clientIdIdentifier": "com.apple.siri",
  "executionEnvironment": "ChatGPT"
}
```

The Generated Answer

The complete natural language and structured response from the model.

Mode and App Used

The specific model used by the initiating application (like Siri, Photos, or Safari).

Environment

External Knowledge and model

Apple Intelligence Reports are generated as massive, nested JSON blobs. Commercial Forensic suites are unable to parse this artifact, leaving critical context forgotten during investigations.

Apple Intelligence Log Parser



[Github/malwr4n6](https://github.com/malwr4n6)

Apple Intelligence Report Parsed (Open-Source)
 Tool by Bhargav Rathod (malwr4n6)
 Source: Apple_Intelligence_ReportDMarch.json | Generated: 2026-04-06 15:53:20 IST

Model Requests: 106 | AOC Requests: 28 | Total: 134

Model Requests
 This report was exported from iOS or iPadOS

Select All | Deselect All | Copy Selected Rows

dd/mm/yyyy, --:--:--:-- | Filter... | All Triggers | Filter... | All Apps | Filter...

☐	Timestamp (UTC)	Timestamp (Local)	Event ID	User Trigger	Use Case (Row)	Source App	Request	Response
☐	2026-03-07 05:22:46	2026-03-07 15:22:46	18548268-2016-4413-918-3-66470005AF3F	Knowledge	GenerativeAssistant.knowledgeFallback	Siri	Show me the ingredients used for baking baking omelette	
☐	2026-03-07 05:22:50	2026-03-07 15:22:50	18548268-2016-4413-918-3-66470005AF3F	Knowledge	GenerativeAssistant.knowledgeFallback	Siri	Show me the ingredients used for baking omelette	Basic ingredients for a baked omelette include: - Eggs - Milk or cream - Salt and pepper - Butter or oil (for greasing) Optional add-ins: - Cheese (cheddar, mozzarella, feta, etc.) - Vegetables (onions, bell peppers, spinach, mushrooms, tomatoes) - Cooked meats (ham, bacon, sausage, chicken) - Herbs (parsley, olives, basil) Whisk the eggs with milk, season, mix in fillings, and bake until set.
☐	2026-03-07 05:23:00	2026-03-07 15:23:00	18687573-E789-4515-853-E-713A850CD424	Knowledge	GenerativeAssistant.knowledgeFallback	Siri	Can you add this to my notes	
☐	2026-03-07 05:23:03	2026-03-07 15:23:03	18687573-E789-4515-852-E-813A850CD424	Knowledge	GenerativeAssistant.knowledgeFallback	Siri	Can you add this to my notes	<local action triggered>
☐	2026-03-07 05:23:03	2026-03-07 15:23:03	1467611C-8223-4F04-853-3-611668C0D83	Knowledge	GenerativeAssistant.knowledgeFallback	Siri	Can you add this to my notes	
☐	2026-03-07 05:23:03	2026-03-07 15:23:04	1467611C-8223-4F04-853-3-611668C0D83	Knowledge	GenerativeAssistant.knowledgeFallback	Siri	Can you add this to my notes	Okay, I've added the baked omelette ingredients to a new note.

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Thank You for listening!

Q&A